INCH-POUND

MIL-M-28787/420
24 October 1991

# MILITARY SPECIFICATION

## MODULES, STANDARD ELECTRONIC

## CONVERTER, D/A MODULE, KEY CODE JFK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic D/A converter module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.619978 failures per million operating hours for class I or 1.83347 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be JFK.

3.6.2 Size. The module size shall be 1A Format A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned 1 through 40.



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TABLE I. Module electrical requirements.

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | Class I and II 25 ± 5°C 4/ | | Class I 0 ± 5°C 4/ 60 ± 5°C | | Class II -55 ± 5°C 4/ 85 ± 5°C | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| A1 | Isolation initial | II | S | 10 | INF | 10 | INF | 10 | INF | MOhm |
| | end-of-life | | S | 1 | INF | 1 | INF | 1 | INF | MOhm |
| A2-A11 | Isolation initial | II | X | 10 | INF | 10 | INF | 10 | INF | MOhm |
| | end-of-life | | X | 1 | INF | 1 | INF | 1 | INF | MOhm |
| B1, B2 | Continuity | III | X | 0 | 1 | 0 | 1 | 0 | 1 | Ohm |
| C | Voo stress | IV | S | ---------- Module shall pass all remaining tests ---------- | | | | | | |
| D | Ioo | V | S | -40 | 0 | -40 | 0 | -40 | 0 | mA |
| E1-E17 | Iih | VI | X | 0 | 300 | 0 | 300 | 0 | 300 | uA |
| F1-F17 | Iil | VII | X | 0 | 100 | 0 | 100 | 0 | 100 | uA |
| G1, G2 | PSFC | VIII | S | 0.75 | 1.25 | 0.75 | 1.25 | 0.75 | 1.25 | uF |
| H1 | Ioo | IX | X | 1 | 8.5 | 1 | 8.5 | 1 | 8.5 | mA |
| H2 | Icc | IX | X | -1 | -25 | -1 | -25 | -1 | -25 | mA |
| J | Resistance | X | X | 4.46 | 4.94 | 4.46 | 4.94 | 4.46 | 4.94 | KOhm |
| K1-K9 | Bipolar accuracy | XI | X | ---------- See table XI ---------- | | | | | | |
| L1-L6 | Unipolar accuracy | XII | X | ---------- See table XII ---------- | | | | | | |
| M1-M32 | Integral linearity | XIII | X | -0.5 | 0.5 | -0.5 | 0.5 | -0.5 | 0.5 | LSB |

- See footnotes at end of table.



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TABLE I. Module electrical requirements - Continued.

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | Class I and II 25 ± 5°C 4/ | | Class I 0 ± 5°C 4/ 60 ± 5°C | | Class II -55 ± 5°C 4/ 85 ± 5°C | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| N1-N15 | Differential linearity | XIV | X | -0.75 | 0.75 | -0.75 | 0.75 | -0.75 | 0.75 | LSB |
| P1 | Power supply sensitivity | XV | S | -3.9 | 3.9 | -4.5 | 4.5 | -4.5 | 4.5 | mV |
| P2 | Power supply sensitivity | XV | S | -2.4 | 2.4 | -3.0 | 3.0 | -3.0 | 3.0 | mV |
| P3 | Power supply sensitivity | XV | S | -3.9 | 3.9 | -4.5 | 4.5 | -4.5 | 4.5 | mV |
| P4 | Power supply sensitivity | XV | S | -2.4 | 2.4 | -3.0 | 3.0 | -3.0 | 3.0 | mV |
| Q1, Q2 | Settling time | XVI | S | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | uS |
| R1 | Timing | XVII | S | -0.5 | 0.5 | -0.5 | 0.5 | -0.5 | 0.5 | V |
| R2 | Timing | XVII | S | -11.0 | -9.0 | -11.0 | -9.0 | -11.0 | -9.0 | V |

- See footnotes on next page.



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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.



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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/420-1 for class I and M28787/420-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram and pin assignments shall conform to figure 1.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for power supply voltages of 15 V dc (Vee) and -15 V dc (-Vee) except that the limits for output short circuit current apply only for power supply voltages of 16.5 V dc (Vee) and -16.5 V dc (-Vee).

3.11 Parallel outputs. Paralleling of outputs is not permissible.



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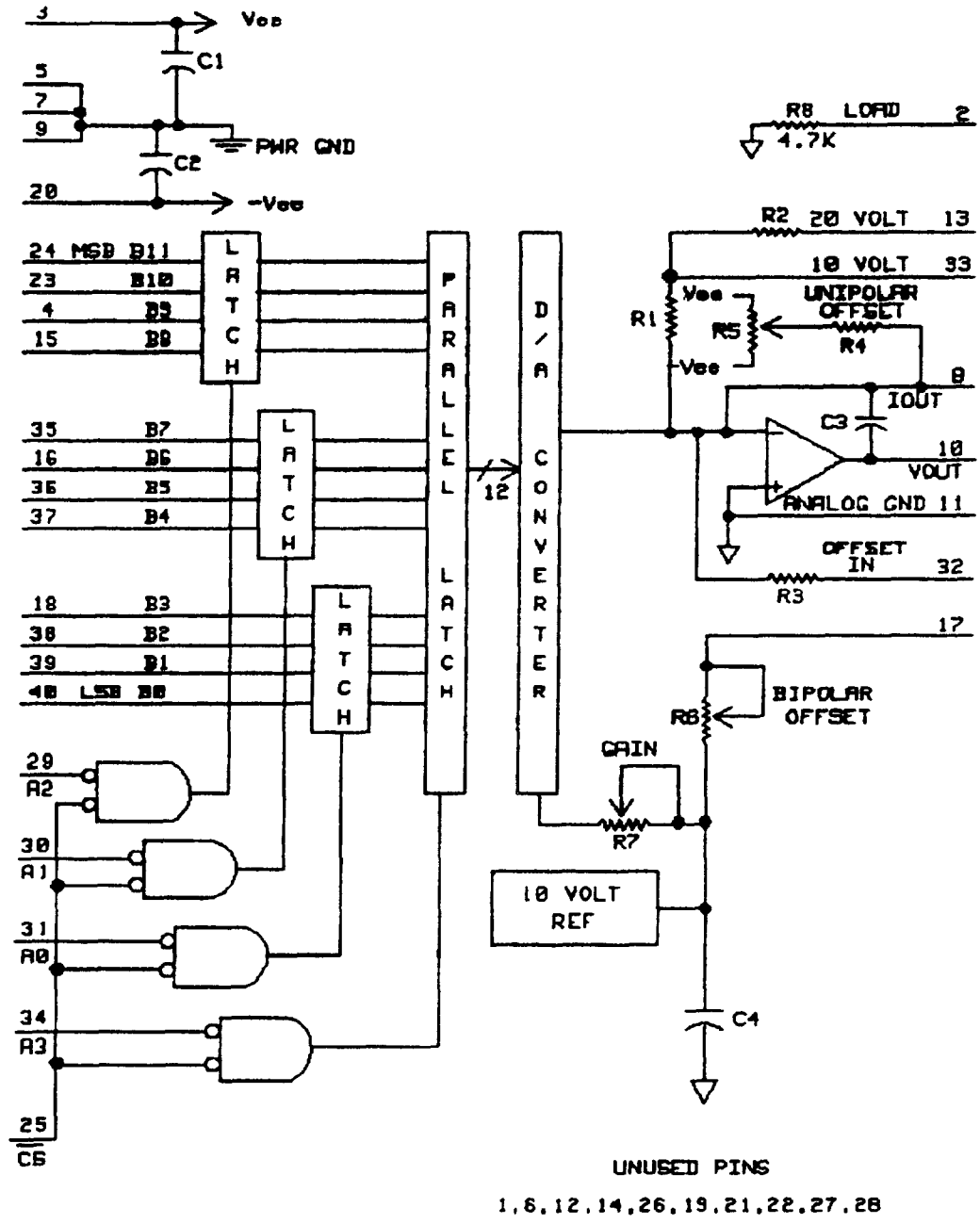



FIGURE 1. JFK functional block diagram and pin assignment.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. Test conditions A, C and D are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C or D are part of a sequence of tests, the order of testing shall be as follows: A, C, and D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B and E through R shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.



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4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition G, shall be performed using the initial limits at 25 ± 5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B and E through R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B and E through R shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

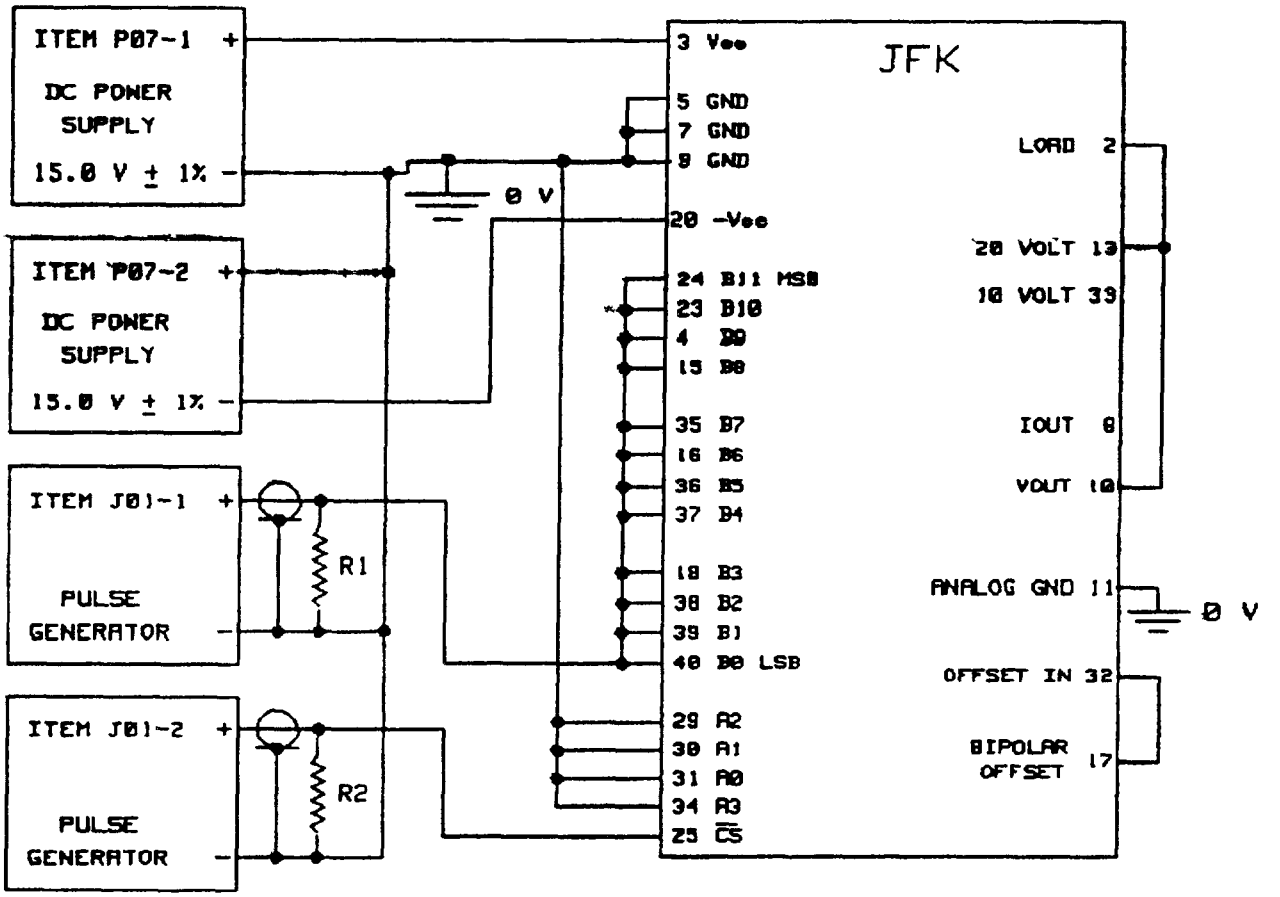
4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.



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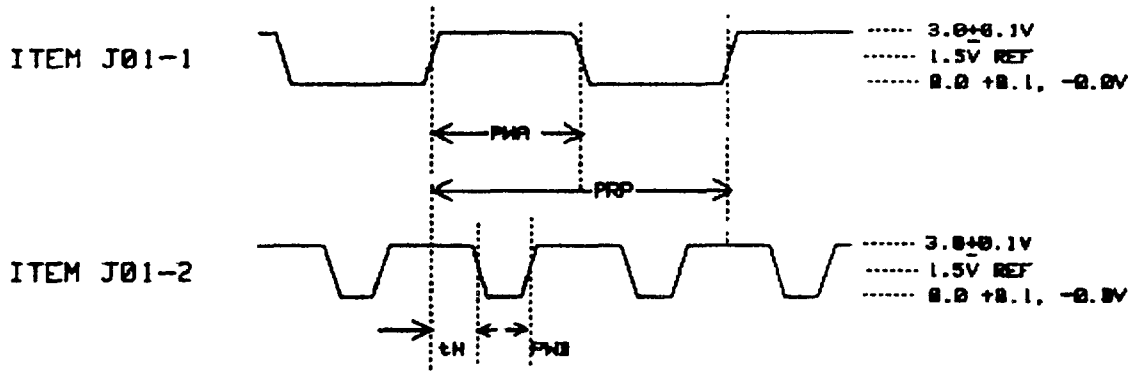

NOTES:

1. See figure 3 for pulse generator setup.
2. R1 = R2 = 50 Ohms ± 1 percent, 1W.

FIGURE 2. JFK operating temperature, component temperature, and life test circuit.



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NOTES:

1. tTLH ≤ 20 nS (10 percent of value to 90 percent)
2. tTHL ≤ 20 nS (90 percent of value to 10 percent)
3. PWA = 10 ± 1 uS
4. PRP = 20 ± 2 uS
5. tH = 2 ± 0.5 uS
6. PWB = 2 ± 0.5 uS

FIGURE 3. Timing diagram for operating temperature, component temperature, and life test.



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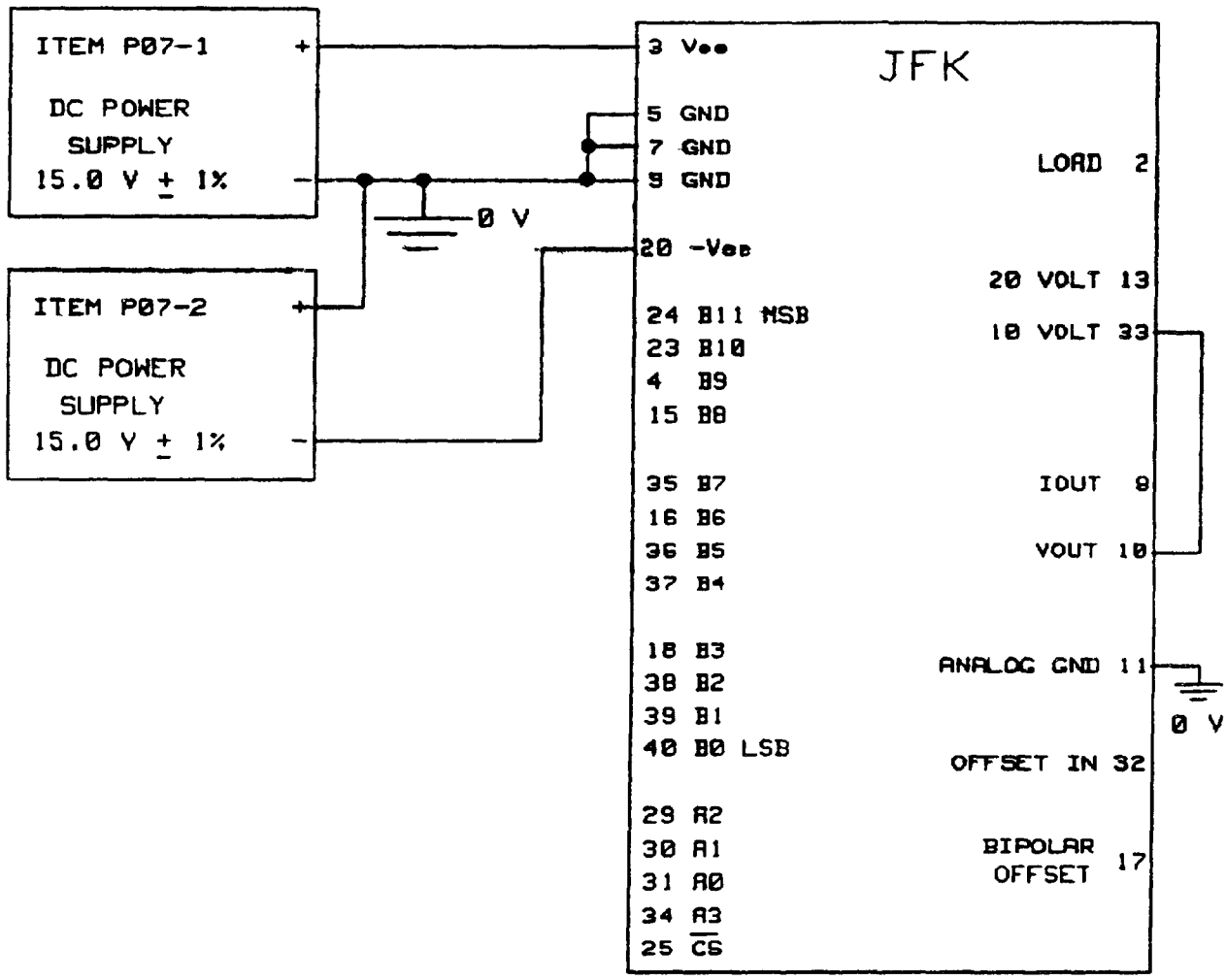



FIGURE 4. JFK vibration test circuit.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Icc = power supply current.
(b) Iih = input high level current.
(c) Iil = input low level current.
(d) Ioh = output high level current.
(e) Iol = output low level current.
(f) Ios = output short circuit current.
(g) LSB = least significant bit.
(h) MSB = most significant bit.
(i) NA = not applicable.
(j) PSFC = power supply filter capacitance.
(k) Vih = input high level voltage.
(l) Vil = input low level voltage.
(m) Voh = output high level voltage.
(n) Vol = output low level voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage ranges: 13.5 V to 16.5 V dc (Vee) and -16.5 V to -13.5 V dc (-Vee).



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6.3.1 Input parameters.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil min (mA) | Iih max (mA) | Ibd max (mA) | Cin max (pF) |
|---|---|---|---|---|---|---|---|
| A0 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| A1 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| A2 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| A3 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| $\overline{CS}$ | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| BIT 11 (MSB) | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| BIT 10 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| BIT 9 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| BIT 8 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| BIT 7 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| BIT 6 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| BIT 5 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| BIT 4 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| BIT 3 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |



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6.3.1 Input parameters - Continued.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil min (mA) | Iih max (mA) | Ibd max (mA) | Cin max (pF) |
|---|---|---|---|---|---|---|---|
| BIT 2 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| BIT 1 | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |
| BIT 0 (LSB) | --- | 0.7 | 2.0 | 0 | 0.3 | --- | --- |

6.4 Module function. The JFK is an 8- or 12-bit digital to analog converter which can be operated in two modes: bipolar, with ranges of -2.5 to +2.5 volts, -5 to +5 volts, and -10 to +10 volts; and unipolar, with ranges of 0 to +5 volts and 0 to +10 volts (see 6.4.1). The nibble input latches, controlled by A0, A1 and A2, allow the JFK to operate on 4-, 8-, or 12-bit busses. The JFK also contains a load resistor equal to 4.7 KOhms.

6.4.1 Mode and range selection.

| Mode | Range | Connect pin 10 to pin | Connect pin 8 to pin | Connect pin 17 to pin |
|---|---|---|---|---|
| Bipolar | ± 2.5 V | 33 | 13 | 32 |
| Bipolar | ± 5 V | 33 and 13 | NC | 32 |
| Bipolar | ± 10 V | 13 | NC | 32 |
| Unipolar | 0 to 5 V | 33 | 13 | NC |
| Unipolar | 0 to 10 V | 33 and 13 | NC | NC |



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6.4.2 Functional description of I/O pins.

| Functional designation | Pin number | I/O | Functional description |
|---|---|---|---|
| A0 | 31 | I | Least significant nibble enable - active low. |
| A1 | 30 | I | Middle significant nibble enable - active low. |
| A2 | 29 | I | Most significant nibble enable - active low. |
| A3 | 34 | I | Parallel latch enable - active low. |
| $\overline{CS}$ | 25 | I | Chip select - active low. |
| BIT 11 (MSB) | 24 | I | Digital input bits. |
| BIT 10 | 23 | I | |
| BIT 9 | 4 | I | |
| BIT 8 | 15 | I | |
| BIT 7 | 35 | I | |
| BIT 6 | 16 | I | |
| BIT 5 | 36 | I | |
| BIT 4 | 37 | I | |
| BIT 3 | 18 | I | |
| BIT 2 | 38 | I | |
| BIT 1 | 39 | I | |
| BIT 0 (LSB) | 40 | I | Digital input bits. |
| LOAD | 2 | I | Load resistor. |



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6.4.2 Functional description of I/O pins - Continued.

| Functional designation | Pin number | I/O | Functional description |
|---|---|---|---|
| 10 VOLT FDBK | 33 | I | Feedback connection for output amplifier. |
| 20 VOLT FDBK | 13 | I | Feedback connection for output amplifier. |
| OFFSET IN | 32 | I | Connection point for bipolar mode. |
| OUTPUT | 10 | O | Analog output. |
| CURRENT OUT | 8 | O | Summing junction of output amplifier. |
| BIP OFFSET | 17 | O | This pin connected to pin 32 places the JFK in bipolar mode. |

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0073)

Agent: NW



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TABLE II. Isolation test.

| Test cond. | Test description | Set-up conditions: Connect A to | Set-up conditions: Connect B to | 25 ± 5°C Initial limits: Min | 25 ± 5°C Initial limits: Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| A1 | Frame | All pins | Frame | 10 | INF | MOhm | 1-8 |
| A2 | Used to unused | All used | All unused | | | | |
| A3 | Unused to unused | 1 | 6,12,14, 26,19, 21,22, 27,28 | | | | |
| A4 | | 6 | 12,14, 26,19, 21,22, 27,28 | | | | |
| A5 | | 12 | 14,26, 19,21, 22,27, 28 | | | | |
| A6 | | 14 | 26,19, 21,22, 27,28 | | | | |
| A7 | | 26 | 19,21, 22,27,28 | | | | |
| A8 | | 19 | 21,22, 27,28 | | | | |
| A9 | | 21 | 22,27,28 | | | | |
| A10 | | 22 | 27,28 | | | | |
| A11 | Unused to unused | 27 | 28 | 10 | INF | MOhm | 1-8 |



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NOTES:

1. Isolation test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Megohm bridge voltage | 100 V<br>30 V | ± 10 percent<br>± 10 percent | A1<br>A2-A11 |
| Isolation measurement | NA | ± 500 KOhm | A1-A11 |
| Isolation measurement (See note 6) | NA | ± 50 KOhm | A1-A11 |

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge for test condition A1.

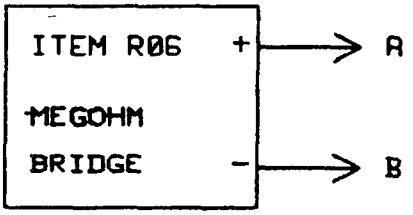
6. The minimum end-of-life limit is 1 MOhm.

7. The test circuit shall be as indicated on figure 5.

8. Abbreviations used are defined in 6.1.



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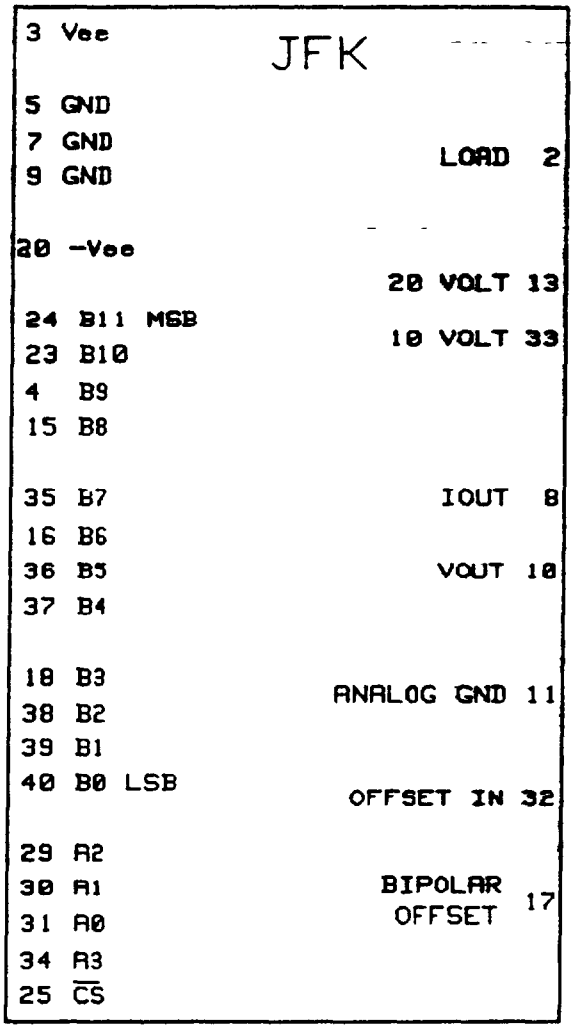




FIGURE 5. Isolation test circuit for table II.



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TABLE III. Continuity test.

| Test cond. | Test description | Set-up conditions Connect A to | Set-up conditions Connect B to | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| B1 | 15 V RETURN | 5 | 9 | 0 | 1 | Ohm | 1-5 |
| B2 | 15 V RETURN | 7 | 9 | 0 | 1 | Ohm | 1-5 |

NOTES:

1. Continuity test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Continuity measurement | NA | ± 0.05 Ohm | B1, B2 |

2. The ohmmeter used shall be item R10.

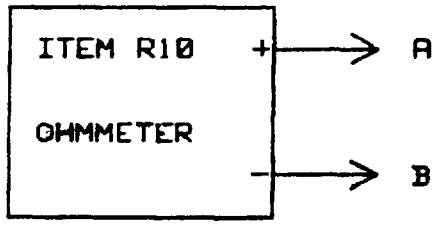
3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 6.

5. Abbreviations used are defined in 6.1.



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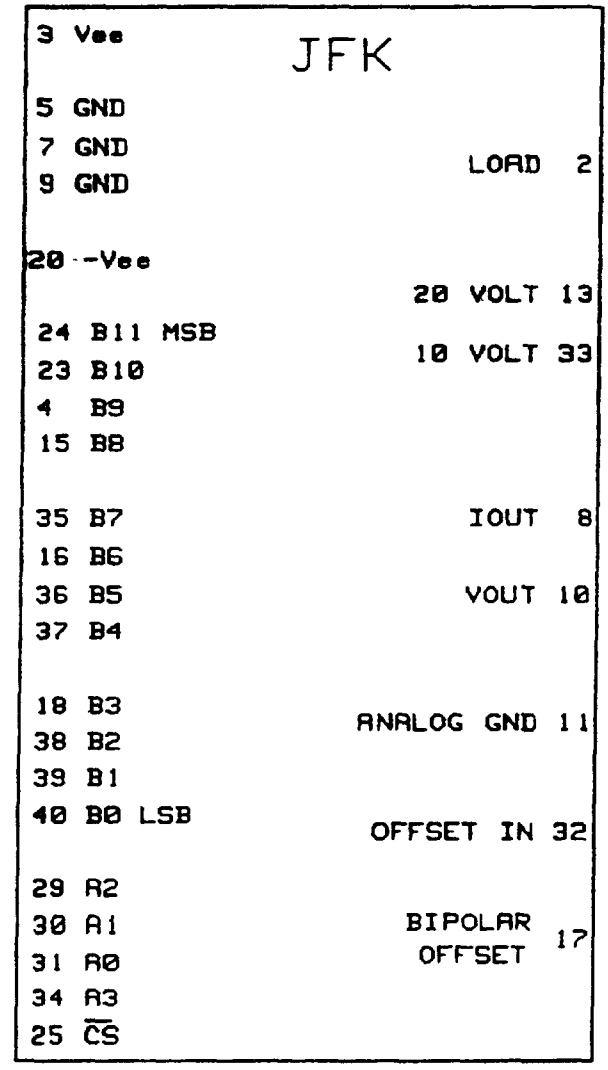




FIGURE 6. Continuity test circuit for table III.



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TABLE IV. Power supply stress test.

| Test cond. | Test description | Set-up conditions Connect A to | Set-up conditions Connect B to | 25 ± 5°C Initial limits | Notes |
|---|---|---|---|---|---|
| C | Power supply stress test | NA | NA | Module shall pass the remaining tests. | 1-5 |

NOTES:

1. Power supply stress test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vee) | 18.0 V | + 0 mV, -180 mV | C |
| (-Vee) | -18.0 V | + 180 mV, -0 mV | C |
| Power supply current limit (Vee) | 15.3 mA | NA | C |
| (-Vee) | -45 mA | NA | C |

2. There shall be no power supply voltage overshoot on this test.

3. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.



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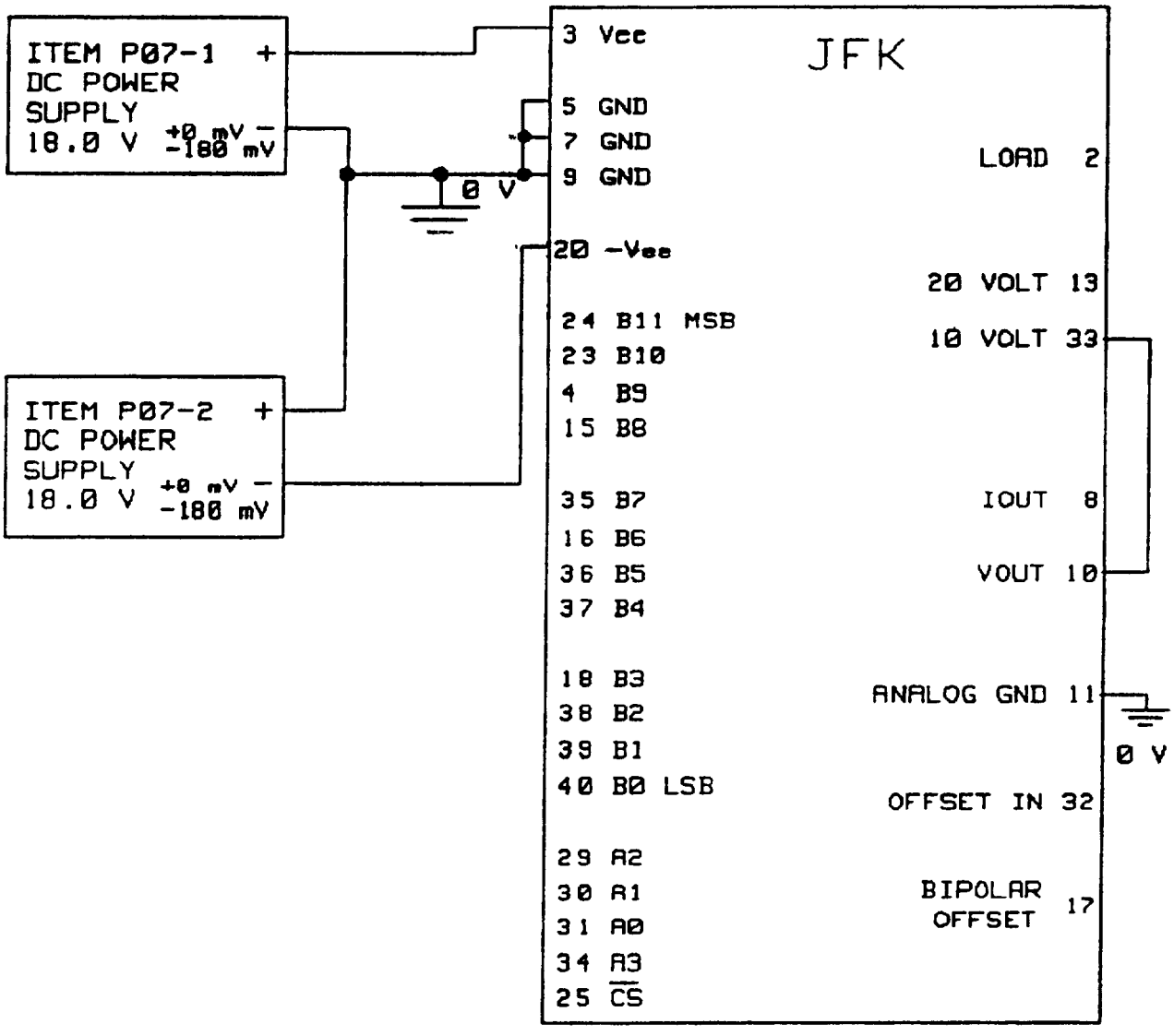



FIGURE 7. Power supply stress test circuit for table IV.



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TABLE V. Output short circuit current test.

| Test cond. | Signal name | Pin no. | Vout (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| D | OUTPUT | 10 | 0.0 | -40 | 0 | mA | 1-5 |

NOTES:

1. Output short circuit current test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vee)<br>(-Vee) | 16.5 V<br>-16.5 V | ± 165 mV<br>± 165 mV | D<br>D |
| Power supply current limit (Vee)<br>(-Vee) | 12.8 mA<br>-37.5 mA | NA<br>NA | D<br>D |
| Forcing voltage, (Vout) | 0.0 V | ± 20 mV | D |
| Pattern voltage high, (VH)<br>low, (VL) | 3.0 V<br>0.0 V | ± 300 mV<br>± 300 mV | D<br>D |
| Output current measurement | NA | ± 2 mA | D |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage overshoot shall not exceed the value specified in the power supply stress test tolerance table.



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NOTES: - Continued.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, the duration of momentarily connecting point A shall not exceed 1 second.

4. The test circuit shall be as indicated on figure 8.

5. Abbreviations used are defined in 6.1.



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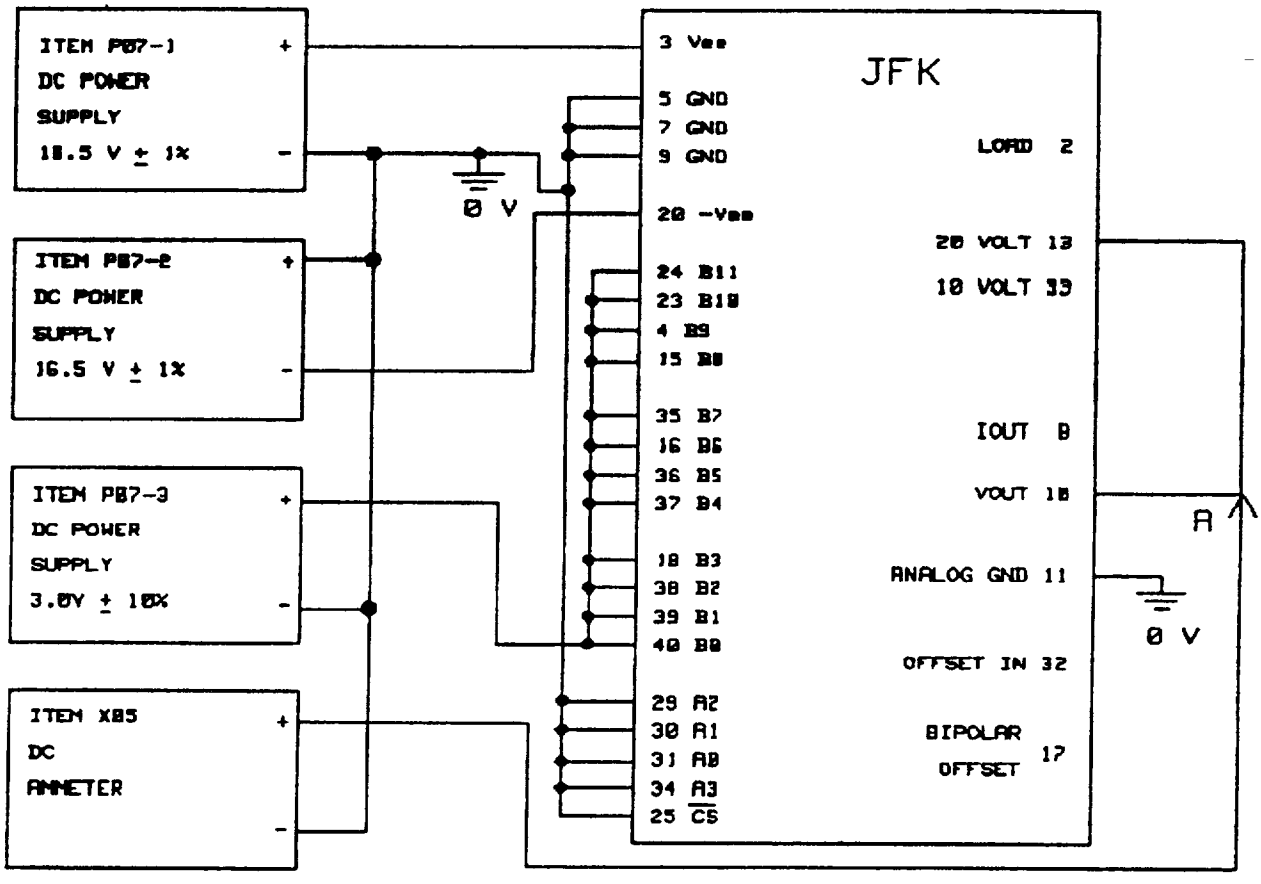



FIGURE 8. Output short circuit test circuit for table V.



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TABLE VI. Input current, high level, test.

| Test cond. | Signal name | Connect A to Pin | Connect B to Pin | Vih (V) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|---|
| E1 | A0 | 31 | OPEN | 5.5 | 0 | 300 | uA | 1-4 |
| E2 | A1 | 30 | | | | | | |
| E3 | A2 | 29 | | | | | | |
| E4 | A3 | 34 | | | | | | |
| E5 | $\overline{CS}$ | 25 | OPEN | | | | | |
| E6 | BIT 11 (MSB) | 24 | 31,30, 29,34, 25 | | | | | |
| E7 | BIT 10 | 23 | | | | | | |
| E8 | BIT 9 | 4 | | | | | | |
| E9 | BIT 8 | 15 | | | | | | |
| E10 | BIT 7 | 35 | | | | | | |
| E11 | BIT 6 | 16 | | | | | | |
| E12 | BIT 5 | 36 | | | | | | |
| E13 | BIT 4 | 37 | | | | | | |
| E14 | BIT 3 | 18 | | | | | | |
| E15 | BIT 2 | 38 | | | | | | |
| E16 | BIT 1 | 39 | | | | | | |
| E17 | BIT 0 (LSB) | 40 | 31,30, 29,34, 25 | 5.5 | 0 | 300 | uA | 1-4 |



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NOTES:

1. Input current, high level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vee) | 16.5 V | ± 165 mV | E1-E17 |
| (-Vee) | -16.5 V | ± 165 mV | E1-E17 |
| Power supply current limit (Vee) | 12.8 mA | NA | E1-E17 |
| (-Vee) | -37.5 mA | NA | E1-E17 |
| Forcing voltage, (Vih) | 5.5 V | ± 275 mV | E1-E17 |
| Input current measurement | NA | ± 15 uA | E1-E17 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage overshoot shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. The test circuit shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.



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TABLE VII. Input current, low level, test.

| Test cond. | Signal name | Connect A to Pin | Connect B to Pin | Vil (V) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|---|
| F1 | A0 | 31 | OPEN | 0.7 | 0 | 100 | uA | 1-4 |
| F2 | A1 | 30 | | | | | | |
| F3 | A2 | 29 | | | | | | |
| F4 | A3 | 34 | | | | | | |
| F5 | $\overline{CS}$ | 25 | OPEN | | | | | |
| F6 | BIT 11 (MSB) | 24 | 31,30, 29,34, 25 | | | | | |
| F7 | BIT 10 | 23 | | | | | | |
| F8 | BIT 9 | 4 | | | | | | |
| F9 | BIT 8 | 15 | | | | | | |
| F10 | BIT 7 | 35 | | | | | | |
| F11 | BIT 6 | 16 | | | | | | |
| F12 | BIT 5 | 36 | | | | | | |
| F13 | BIT 4 | 37 | | | | | | |
| F14 | BIT 3 | 18 | | | | | | |
| F15 | BIT 2 | 38 | | | | | | |
| F16 | BIT 1 | 39 | | | | | | |
| F17 | BIT 0 (LSB) | 40 | 31,30, 29,34, 25 | 0.7 | 0 | 100 | uA | 1-4 |



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NOTES:

1. Input current, low level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vee)<br>(-Vee) | 16.5 V<br>-16.5 V | ± 165 mV<br>± 165 mV | F1-F17<br>F1-F17 |
| Power supply current limit (Vee)<br>(-Vee) | 12.8 mA<br>-37.5 mA | NA<br>NA | F1-F17<br>F1-F17 |
| Forcing voltage, (Vil) | 0.7 V | ± 35 mV | F1-F17 |
| Input current measurement | NA | ± 5 uA | F1-F17 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage overshoot shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

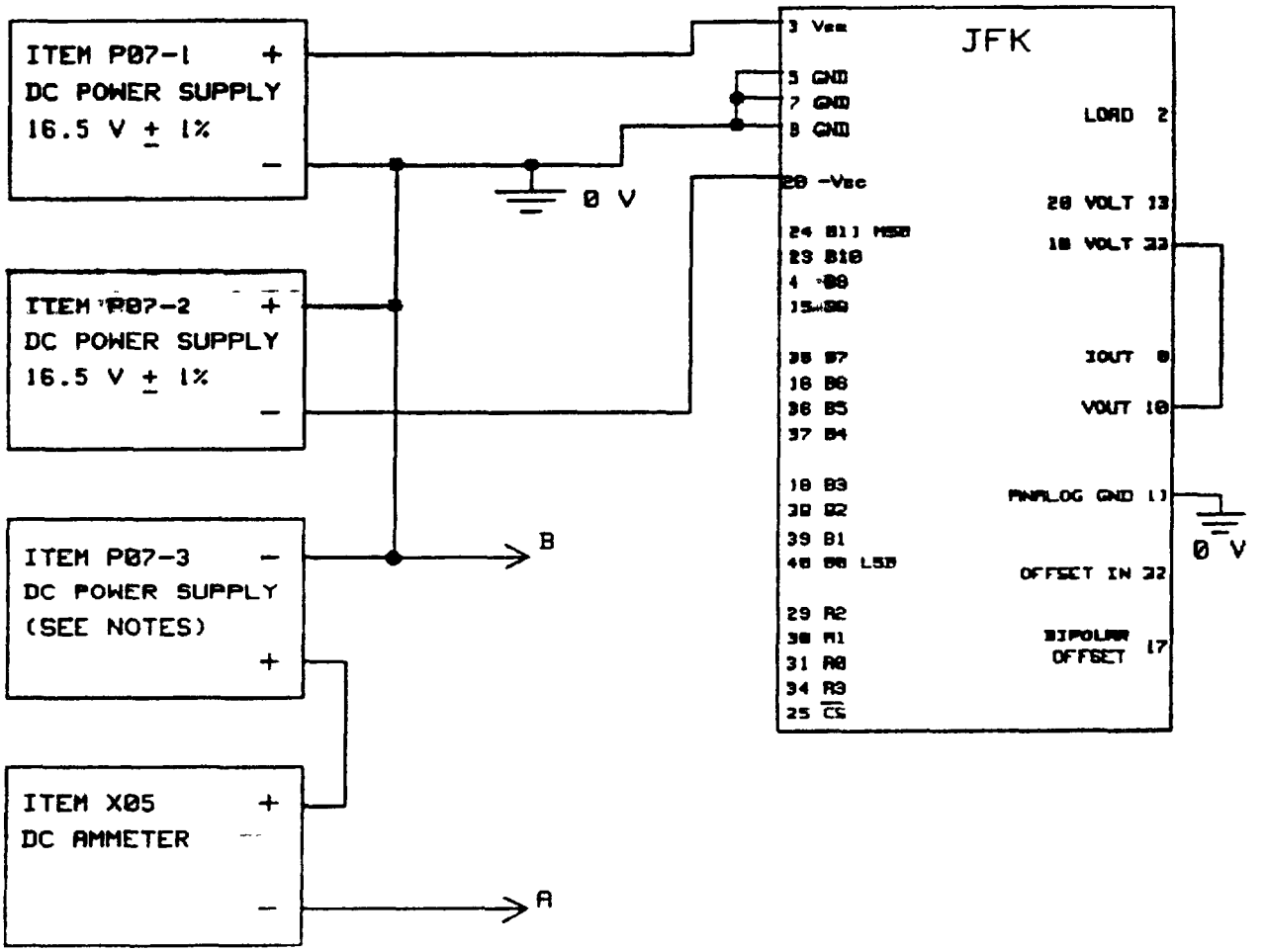
(c) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. The test circuit shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.



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NOTE:

1. Item P07-3 set-up:

5.5 V ± 5 percent for E1 through E17
0.7 V ± 5 percent for F1 through F17

FIGURE 9. Input current test circuit for tables VI and VII.



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TABLE VIII. Power supply filter capacitance test.

| Test cond. | Parameter | Pin numbers | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| G1 | PSFC | 3 | 0.75 | 1.25 | uF | 1-5 |
| G2 | PSFC | 20 | 0.75 | 1.25 | uF | 1-5 |

NOTES:

1. Power supply filter capacitance test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Capacitance bridge frequency | 1 KHz | ± 100 Hz | G1, G2 |
| Capacitance bridge voltage | 0.3 Vp | ± 30 mVp | G1, G2 |
| Capacitance measurement | NA | ± 25 nF | G1, G2 |

2. The capacitance bridge used in this test shall be item C01.

3. For test condition G1, connect the positive terminal to pin 3 and the negative terminal to pin 5. For test condition G2, connect the positive terminal to pin 5 and the negative terminal to pin 20.

4. The test circuit shall be as indicated on figure 10.

5. Abbreviations used are defined in 6.1.



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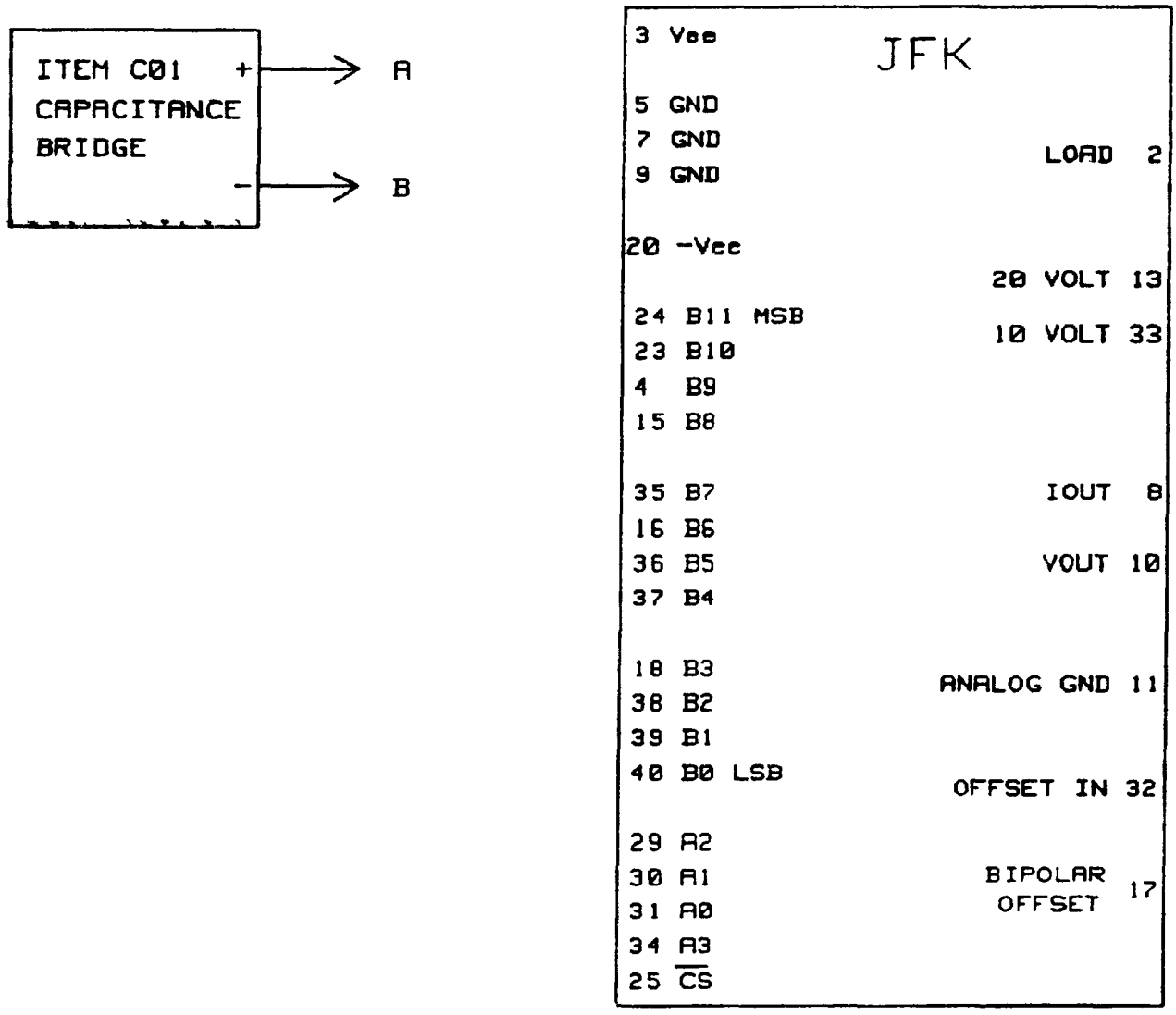



FIGURE 10. Power supply filter capacitance test circuit for table VIII.



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TABLE IX. Power supply current test.

| Test cond. | Signal name | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| H1 | Vee (+15V) | 3 | 1 | 8.5 | mA | 1-4 |
| H2 | -Vee (-15V) | 20 | 1 | -25 | mA | 1-4 |

NOTES:

1. Power supply current test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vee) | 16.5 V | ± 165 mV | H1, H2 |
| (-Vee) | -16.5 V | ± 165 mV | H1, H2 |
| Power supply current limit (Vee) | 12.8 mA | NA | H1, H2 |
| (-Vee) | -37.5 mA | NA | H1, H2 |
| Pattern voltage high, (VH) | 3.0 V | ± 100 mV | H1, H2 |
| low, (VL) | 0.0 V | ± 100 mV | H1, H2 |
| Power supply current measurement | NA | ± 0.38 mA | H1 |
| | NA | ± 0.75 mA | H2 |



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NOTES: - Continued.

2. The following procedure is required to perform this test:

The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage overshoot shall not exceed the value specified in the power supply stress test tolerance table.

3. The test circuit shall be as indicated on figure 11.

4. Abbreviations used are defined in 6.1.



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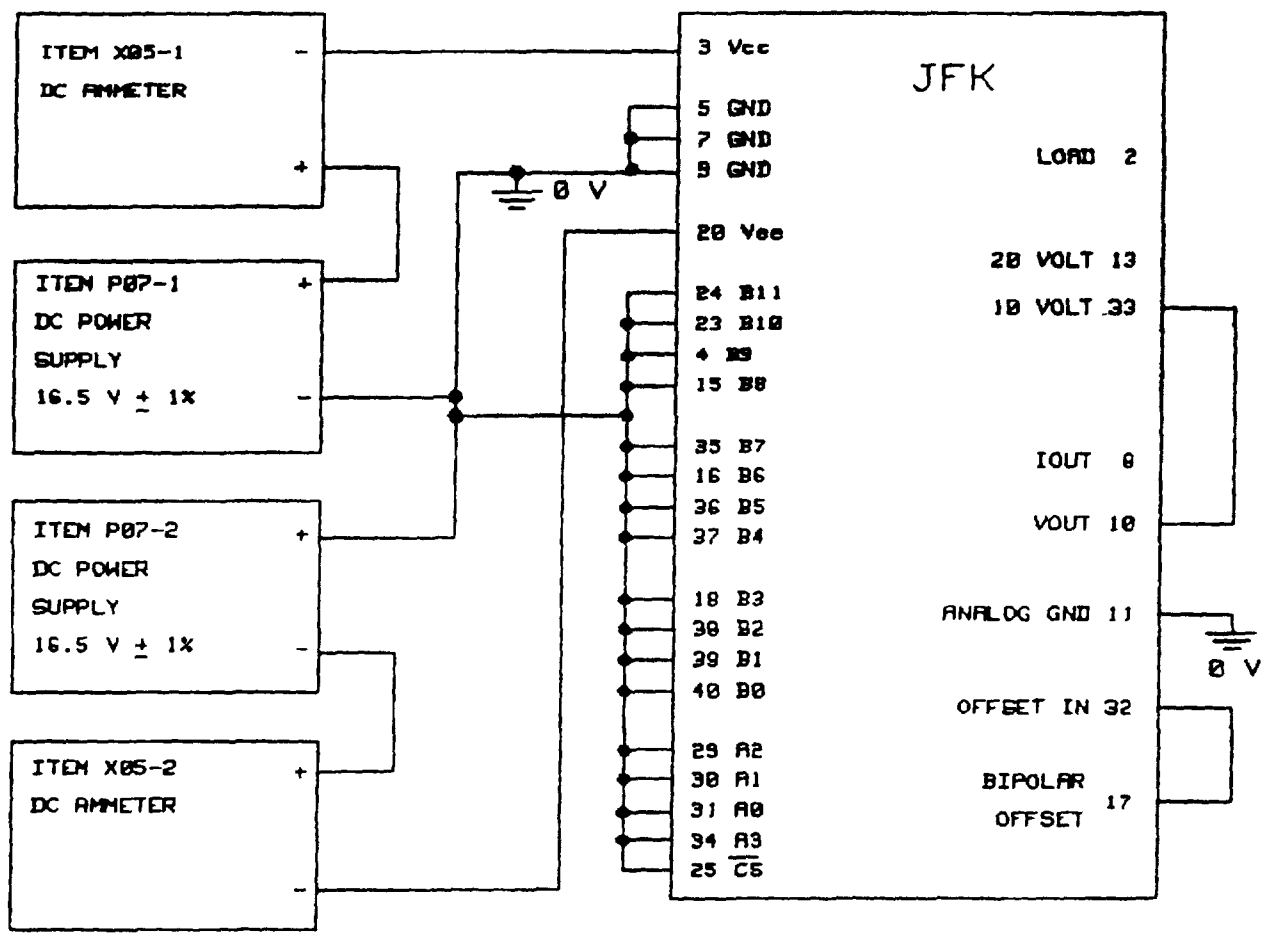



FIGURE 11. Power supply current test circuit for table IX.



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TABLE X. Resistance test.

| Test cond. | Signal name | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| J | LOAD | 2 | 4.46 | 4.94 | KOhm | 1-4 |

NOTES:

1. Resistance test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Digital multimeter range | 10 KOhm | NA | J |
| Resistance measurement | NA | ± 24 Ohm | J |

2. The digital multimeter used in this test shall be item M02. The negative terminal shall be connected to pin 11 and the positive terminal to the pin indicated in table X.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.



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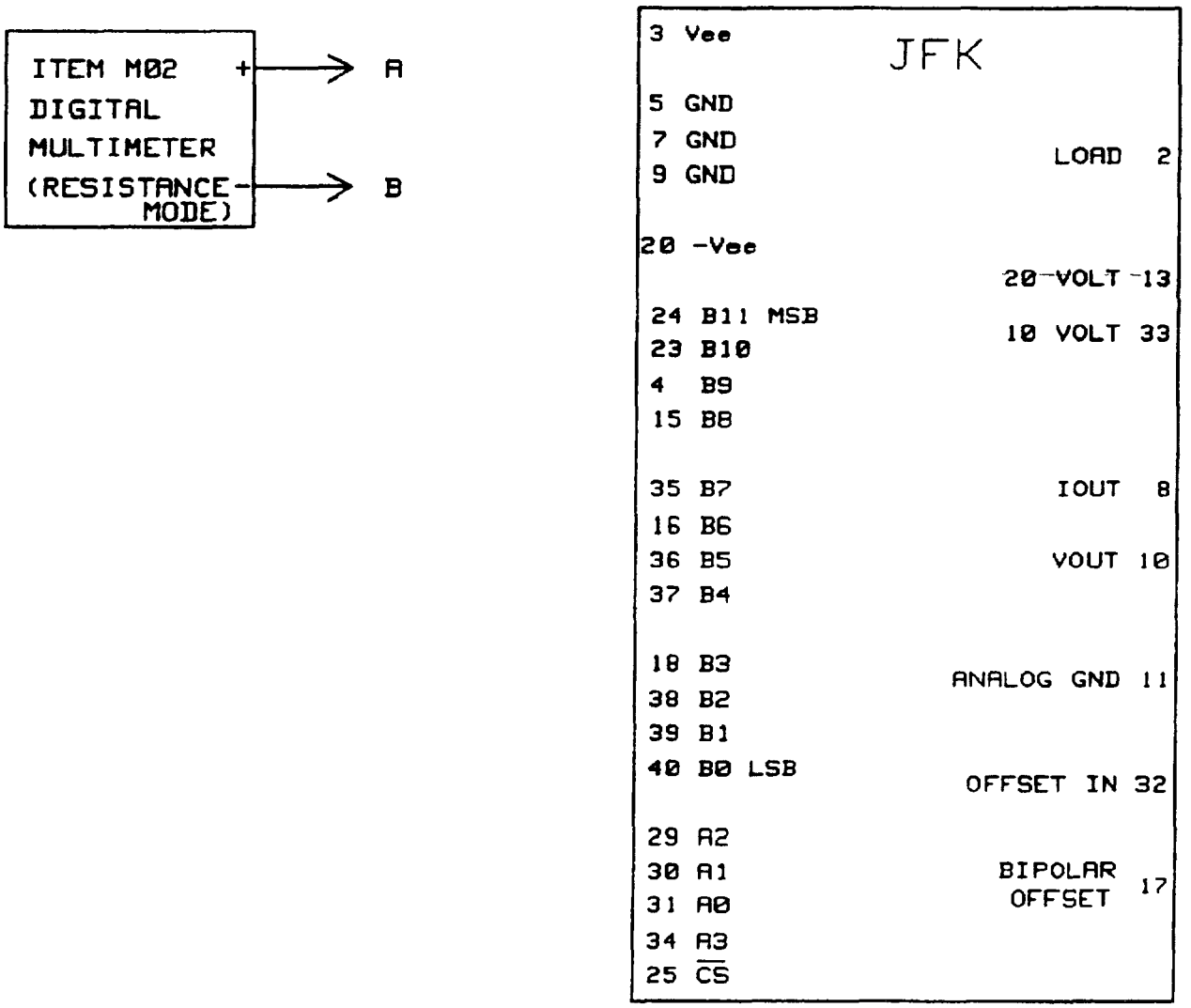



FIGURE 12. Resistance test circuit for table X.



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TABLE XI. Bipolar accuracy test.

| Test cond. | Range | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| K1 | ± 2.5 V | OPEN | 24,23,4, 15,C | -2.5037 | -2.4963 | V | 1-5 |
| K2 | ± 2.5 V | 24 | 23,4,15, C | -0.0037 | 0.0037 | | |
| K3 | ± 2.5 V | 24,23,4, 15,C | OPEN | 2.4951 | 2.5024 | | |
| K4 | ± 5 V | OPEN | 24,23,4, 15,C | -5.0073 | -4.9927 | | |
| K5 | ± 5 V | 24 | 23,4,15, C | -0.0073 | 0.0073 | | |
| K6 | ± 5 V | 24,23,4, 15,C | OPEN | 4.9902 | 5.0049 | | |
| K7 | ± 10 V | OPEN | 24,23,4, 15,C | -10.0146 | -9.9854 | | |
| K8 | ± 10 V | 24 | 23,4,15, C | -0.0146 | 0.0146 | | |
| K9 | ± 10 V | 24,23,4, 15,C | OPEN | 9.9805 | 10.0098 | V | 1-5 |



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NOTES:

1. Bipolar accuracy test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vee) | 13.5 V | ± 135 mV | K1-K9 |
| (-Vee) | -13.5 V | ± 135 mV | K1-K9 |
| Power supply current limit (Vee) | 12.8 mA | NA | K1-K9 |
| (-Vee) | -37.5 mA | NA | K1-K9 |
| Pattern voltage, (VH) | 2.0 V | ± 100 mV | K1-K9 |
| (VL) | 0.7 V | ± 35 mV | K1-K9 |
| Output voltage measurement | NA | ± 0.5 mV | K1-K3 |
| | | ± 1.0 mV | K4-K6 |
| | | ± 2.0 mV | K7-K9 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage overshoot shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the output voltage then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. See 6.4.1 for operating mode setup.

4. The test circuit shall be as indicated on figure 13.

5. Abbreviations used are defined in 6.1.



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TABLE XII. Unipolar accuracy test.

| Test cond. | Range | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| L1 | 0 to +5 V | OPEN | 24,23,4, 15,C | -0.0037 | 0.0037 | V | 1-5 |
| L2 | 0 to +5 V | 24 | 23,4,15, C | 2.4963 | 2.5037 | | |
| L3 | 0 to +5 V | 24,23,4, 15,C | OPEN | 4.9951 | 5.0024 | | |
| L4 | 0 to +10 V | OPEN | 24,23,4, 15,C | -0.0073 | 0.0073 | | |
| L5 | 0 to +10 V | 24 | 23,4,15, C | 4.9927 | 5.0073 | | |
| L6 | 0 to +10 V | 24,23,4, 15,C | OPEN | 9.9902 | 10.0049 | V | 1-5 |

NOTES:

1. Unipolar accuracy test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vee) | 13.5 V | ± 135 mV | L1-L6 |
| (-Vee) | -13.5 V | ± 135 mV | L1-L6 |
| Power supply current limit (Vee) | 12.8 mA | NA | L1-L6 |
| (-Vee) | -37.5 mA | NA | L1-L6 |



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NOTES: - Continued.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Pattern voltage, (VH)<br>(VL) | 2.0 V<br>0.7 V | ± 100 mV<br>± 35 mV | L1-L6<br>L1-L6 |
| Output voltage measurement | NA | ± 0.5 mV<br>± 1.0 mV | L1-L3<br>L4-L6 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage overshoot shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the output voltage then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. See 6.4.1 for operating mode setup.

4. The test circuit shall be as indicated on figure 13.

5. Abbreviations used are defined in 6.1.



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TABLE XIII. Integral linearity test.

| Test cond. | Parameter | Connect A to Pin | Connect B to Pin | n | 25 ± 5°C Initial limit Min | 25 ± 5°C Initial limit Max | Units | Notes |
|---|---|---|---|---|---|---|---|---|
| M1 | Integral linearity | 24, 23,4, 15,C | OPEN | 4095 | -0.5 | 0.5 | LSB | 1-6 |
| M2 | | 24, 23,4, 15 | C | 3840 | | | | |
| M3 | | 24, 23,4, C | 15 | 3839 | | | | |
| M4 | | 24, 23,4 | 15,C | 3584 | | | | |
| M5 | | 24, 23, 15,C | 4 | 3583 | | | | |
| M6 | | 24, 23,15 | 4,C | 3328 | | | | |
| M7 | | 24, 23,C | 4,15 | 3327 | | | | |
| M8 | | 24,23 | 4,15, C | 3072 | | | | |
| M9 | | 24,4, 15,C | 23 | 3071 | | | | |
| M10 | | 24,4, 15 | 23,C | 2816 | | | | |
| M11 | | 24,4, C | 23,15 | 2815 | | | | |
| M12 | Integral linearity | 24,4 | 23,15, C | 2560 | -0.5 | 0.5 | LSB | 1-6 |



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TABLE XIII. Integral linearity test - Continued.

| Test cond. | Parameter | Connect A to Pin | Connect B to Pin | n | 25 ± 5°C Initial limit Min | 25 ± 5°C Initial limit Max | Units | Notes |
|---|---|---|---|---|---|---|---|---|
| M13 | Integral linearity | 24, 15,C | 23,4 | 2559 | -0.5 | 0.5 | LSB | 1-6 |
| M14 | | 24,15 | 23,4, C | 2304 | | | | |
| M15 | | 24,C | 23,4, 15 | 2303 | | | | |
| M16 | | 24 | 23,4, 15,C | 2048 | | | | |
| M17 | | 23,4, 15,C | 24 | 2047 | | | | |
| M18 | | 23,4, 15 | 24,C | 1792 | | | | |
| M19 | | 23,4, C | 24,15 | 1791 | | | | |
| M20 | | 23,4 | 24, 15,C | 1536 | | | | |
| M21 | | 23, 15,C | 24,4 | 1535 | | | | |
| M22 | | 23,15 | 24,4, C | 1280 | | | | |
| M23 | | 23,C | 24,4, 15 | 1279 | | | | |
| M24 | | 23 | 24,4, 15,C | 1024 | | | | |
| M25 | Integral linearity | 4,15, C | 24,23 | 1023 | -0.5 | 0.5 | LSB | 1-6 |



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TABLE XIII. Integral linearity test - Continued.

| Test cond. | Parameter | Connect A to Pin | Connect B to Pin | n | 25 ± 5°C Initial limit Min | 25 ± 5°C Initial limit Max | Units | Notes |
|---|---|---|---|---|---|---|---|---|
| M26 | Integral linearity | 4,15 | 24, 23,C | 768 | -0.5 | 0.5 | LSB | 1-6 |
| M27 | | 4,C | 24, 23,15 | 767 | | | | |
| M28 | | 4 | 24,23, 15,C | 512 | | | | |
| M29 | | 15,C | 24, 23,4 | 511 | | | | |
| M30 | | 15 | 24,23, 4,C | 256 | | | | |
| M31 | | C | 24,23, 4,15 | 255 | | | | |
| M32 | Integral linearity | OPEN | 24,23, 4,15, C | 0 | -0.5 | 0.5 | LSB | 1-6 |



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NOTES:

1. Integral linearity test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vee) | 15.0 V | ± 150 mV | M1-M32 |
| (-Vee) | -15.0 V | ± 150 mV | M1-M32 |
| Power supply current limit (Vee) | 12.8 mA | NA | M1-M32 |
| (-Vee) | -37.5 mA | NA | M1-M32 |
| Forcing voltage, (Vil) | 2.0 V | ± 100 mV | M1-M32 |
| | 0.7 V | ± 35 mV | M1-M32 |
| Output voltage measurement | NA | ± 0.1 mV | M1-M32 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage overshoot shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the output voltage then measured. The forcing voltage overshoot shall not exceed 100 mV.



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NOTES: - Continued.

3. The integral linearity test requires the following calculations:

(a) (M1-M32)/4095 = Average code width, W.

(b) Vm-(nW+M32) = X, the difference between the actual output voltage and the expected output voltage for test condition Mm.

(c) X shall conform to the limits in table XIII.

(d) n is as specified in table XIII.

(e) Vm is the voltage measured for test condition Mm where m is the number of the test condition (For M3, m=3).

4. For test conditions M1-M32, 1LSB = 2.441 mV.

5. The test circuit shall be as indicated on figure 13.

6. Abbreviations used are defined in 6.1.



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TABLE XIV. Differential linearity test.

| Test cond. | Parameter | 25 ± 5°C Initial limit Min | Max | Units | Notes |
|---|---|---|---|---|---|
| N1 | Differential linearity | -0.75 | 0.75 | LSB | 1-4 |
| N2 | | | | | |
| N3 | | | | | |
| N4 | | | | | |
| N5 | | | | | |
| N6 | | | | | |
| N7 | | | | | |
| N8 | | | | | |
| N9 | | | | | |
| N10 | | | | | |
| N11 | | | | | |
| N12 | | | | | |
| N13 | | | | | |
| N14 | | | | | |
| N15 | Differential linearity | -0.75 | 0.75 | LSB | 1-4 |

NOTES:

1. The differential linearity test requires the following calculations:

   (a) Using test conditions M1 and M32, (M1-M32)/4095 = W.

   (b) Vme-Vmo-W shall conform to the limits in table XIV.



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NOTES: - Continued.

(c) Vme is the voltage measured in the even numbered M tests starting with M2 and ending with M30.

(d) Vmo is the voltage measured in the odd numbered M tests starting with M3 and ending with M31.

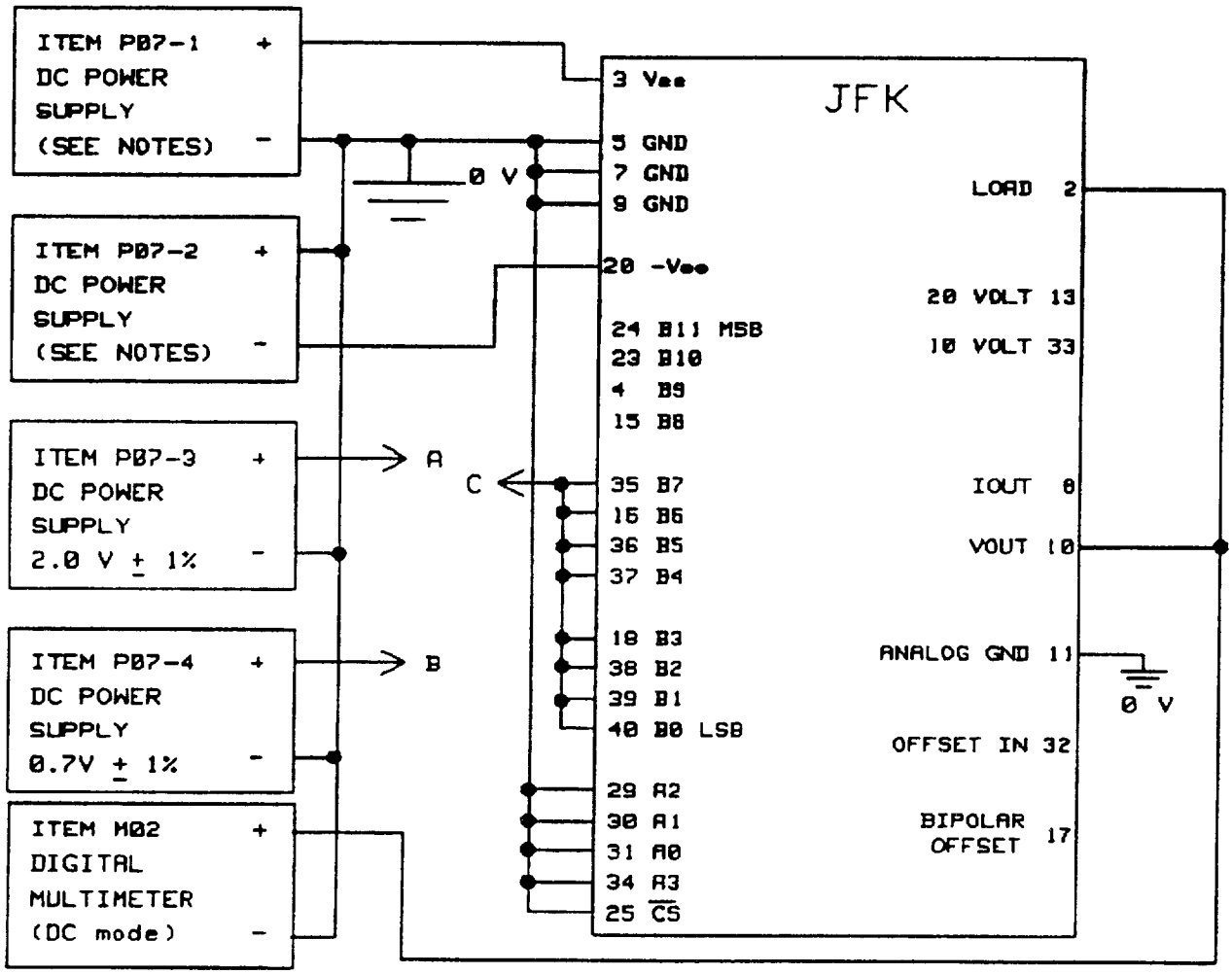
2. For test conditions N1-N15, 1LSB = 2.441 mV.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.



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NOTE:

1. Set up power supplies as per test tolerance table on each test (XI - XIV).

FIGURE 13. Circuit for unipolar and bipolar accuracy tests and linearity tests for tables XI, XII, XIII and XIV.



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TABLE XV. Power supply sensitivity test.

| Test cond. | Signal name | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| P1 | Vee (+15V) | 3 | -3.9 | 3.9 | mV | 1-4 |
| P2 | -Vee (-15V) | 20 | -2.4 | 2.4 | | |
| P3 | Vee (+15V) | 3 | -3.9 | 3.9 | | |
| P4 | -Vee (-15V) | 20 | -2.4 | 2.4 | mV | 1-4 |

NOTES:

1. Power supply sensitivity test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vee) | 16.5 | ± 165 mV | P1 |
| (-Vee) | -15.0 | ± 150 mV | P1 |
| (Vee) | 15.0 | ± 150 mV | P2 |
| (-Vee) | -16.5 | ± 165 mV | P2 |
| (Vee) | 13.5 | ± 135 mV | P3 |
| (-Vee) | -15.0 | ± 150 mV | P3 |
| (Vee) | 15.0 | ± 150 mV | P4 |
| (-Vee) | -13.5 | ± 135 mV | P4 |
| Power supply current limit (Vee) | 12.8 mA | NA | P1-P4 |
| (-Vee) | -37.5 mA | NA | P1-P4 |
| Pattern voltage high, (VH) | 3.0 V | ± 100 mV | P1-P4 |
| low, (VL) | 0.0 V | ± 100 mV | P1-P4 |
| Output voltage measurement | NA | ± 0.1 mV | P1-P4 |



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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) Power supplies Vee and -Vee shall be at nominal voltages and the output voltage noted. The power supplies shall then be adjusted to the values shown in the test tolerance table. The difference between the output voltage with Vee and -Vee at nominal values and the output voltage for each test condition shall conform to the test limits in table XV.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.



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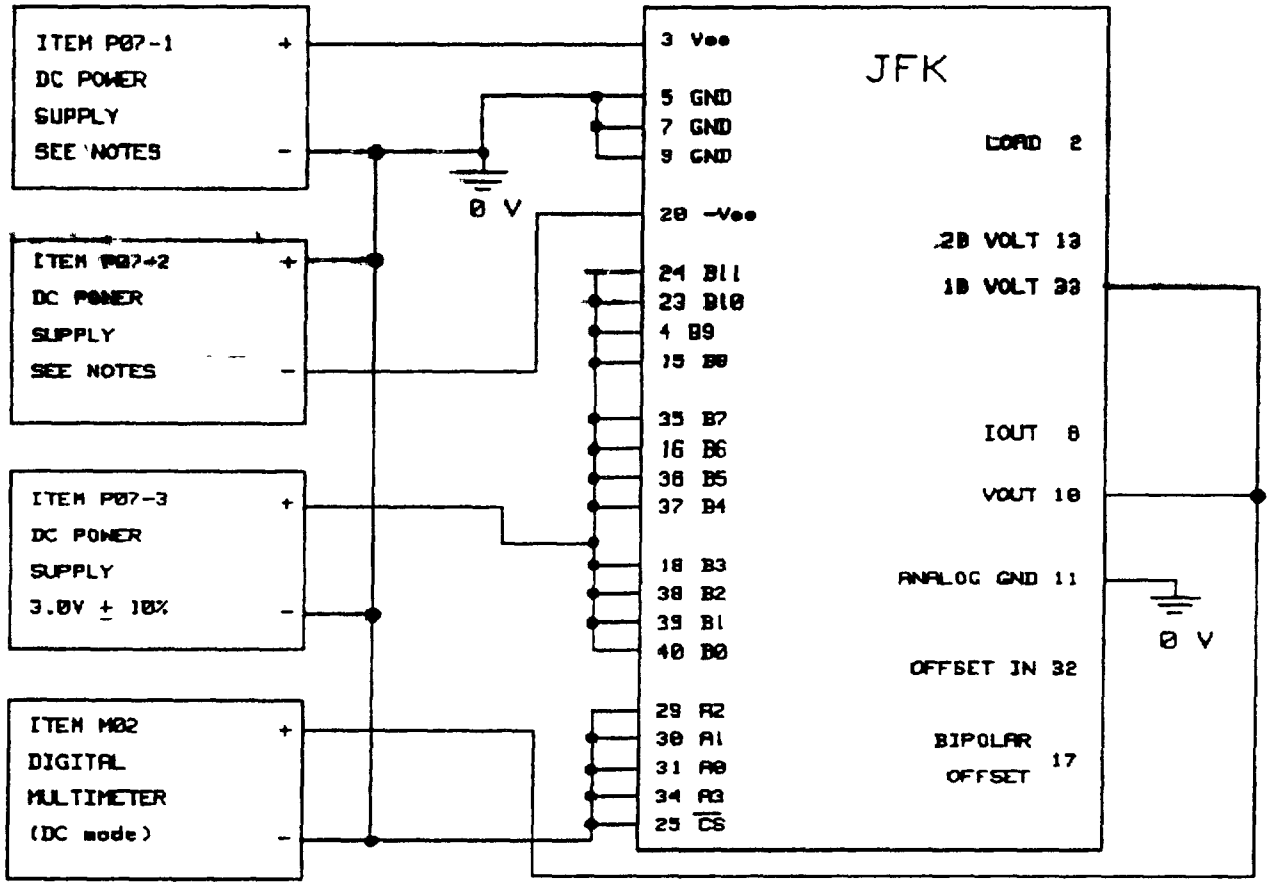



FIGURE 14. Power supply sensitivity test circuit for table XV.



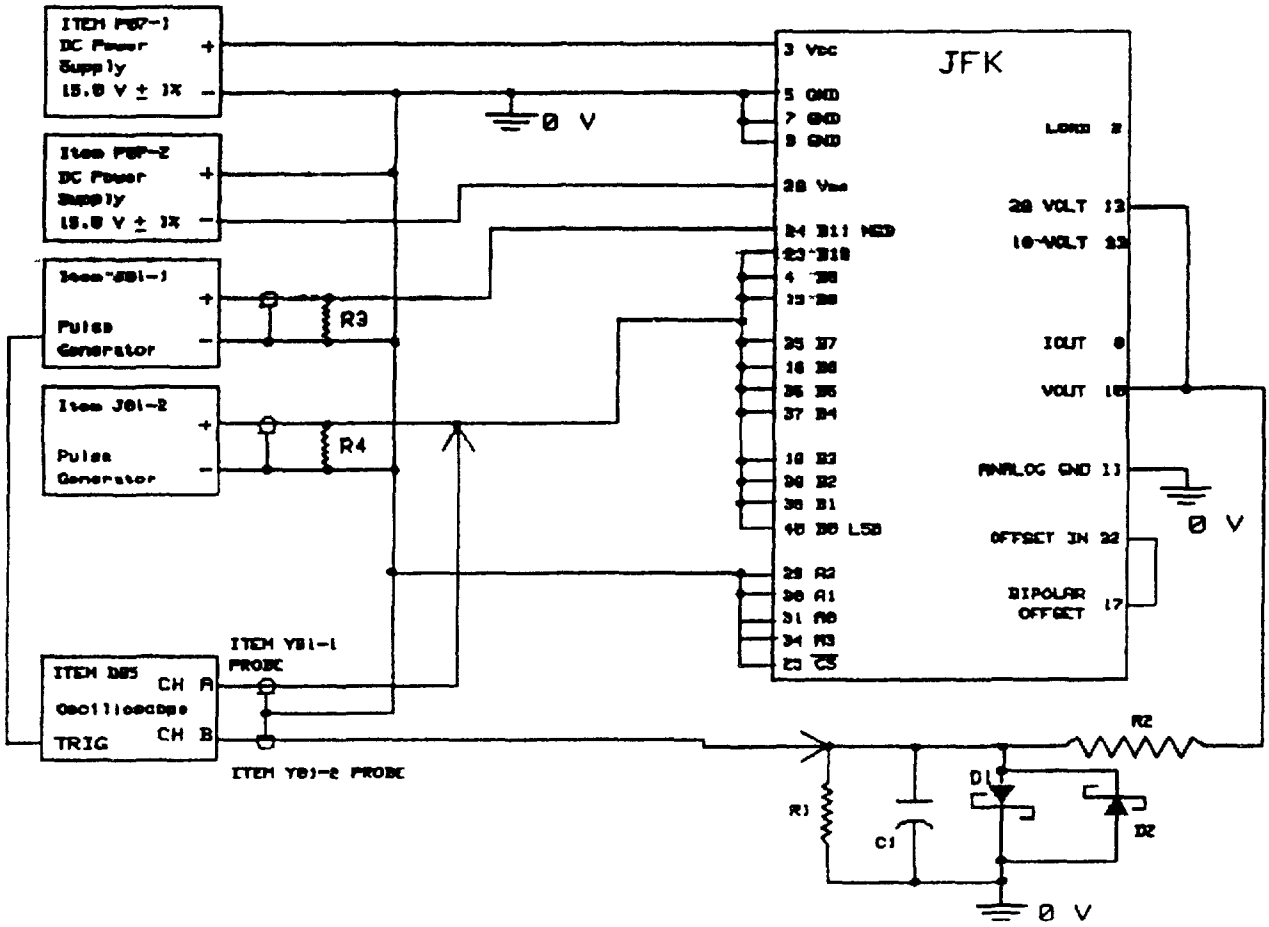
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TABLE XVI. Settling time test.

| Test cond. | Parameter | 25 ± 5°C Initial limits Min | Max | Units | Figs. |
|---|---|---|---|---|---|
| Q1 | Settling time | 0.0 | 8.0 | uS | 15,16 |
| Q2 | Settling time | 0.0 | 8.0 | uS | 15,16 |



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NOTES:

1. R1 = 2 KOhm ± 1 percent, 0.05W.
2. R2 = 10 KOhm ± 1 percent, 0.05W.
3. C1 = 500 pF ± 10 percent.
4. D1 = D2, schottky diodes.
5. R3 = R4 = 50 Ohm ± 1 percent, 0.5W.

FIGURE 15. Settling time test circuit for table XVI.



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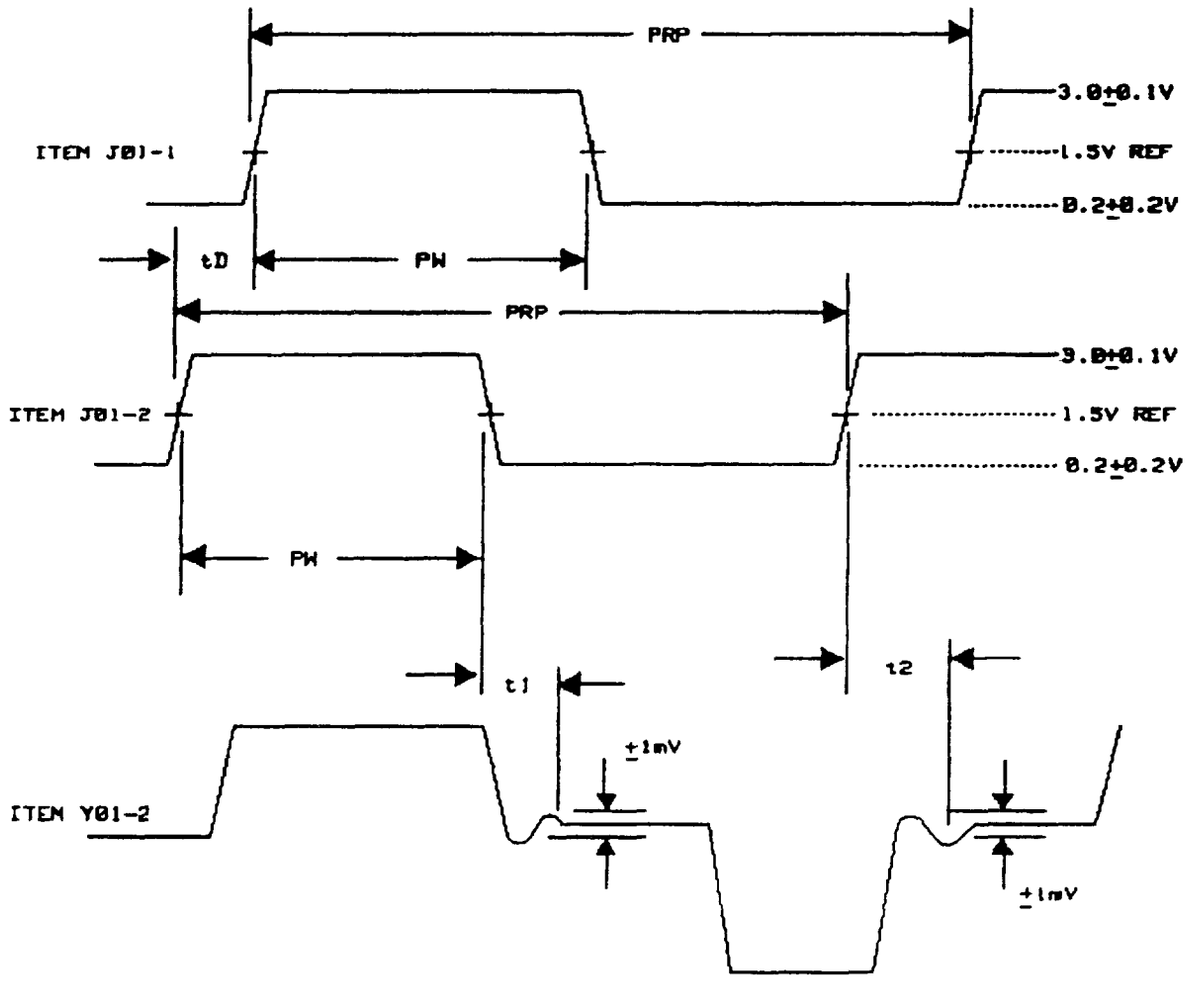



FIGURE 16. Dynamic waveforms for table XVI.



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NOTES:

1. PRP = 80 uS ± 8 uS

2. PW = 40 uS ± 4 uS

3. tD = 20 uS ± 2 uS

4. tTLH = tTHL = 50 nS max, (10 percent to 90 percent)

5. The setting time is defined as the time required for the output to settle within 1 mV of its final value.

6. For test condition Q1, measure t1.

7. For test condition Q2, measure t2.



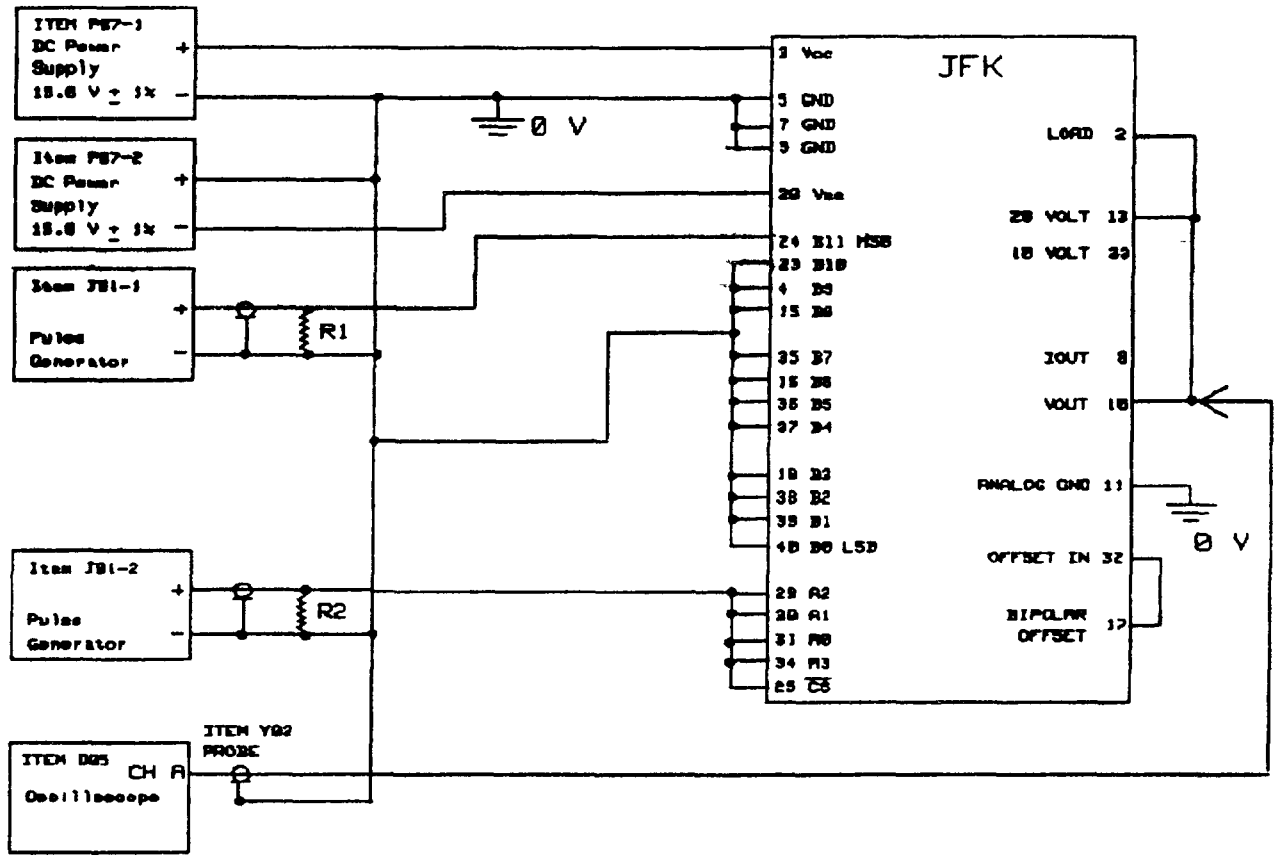
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TABLE XVII. Timing test.

| Test cond. | Parameter | 25 ± 5°C Initial limits Min | Max | Units | Figs. |
|---|---|---|---|---|---|
| R1 | Timing (VOH) | -0.5 | 0.5 | Vdc | 17,18 |
| R2 | Timing (VOL) | -11.0 | -9.0 | Vdc | 17,18 |



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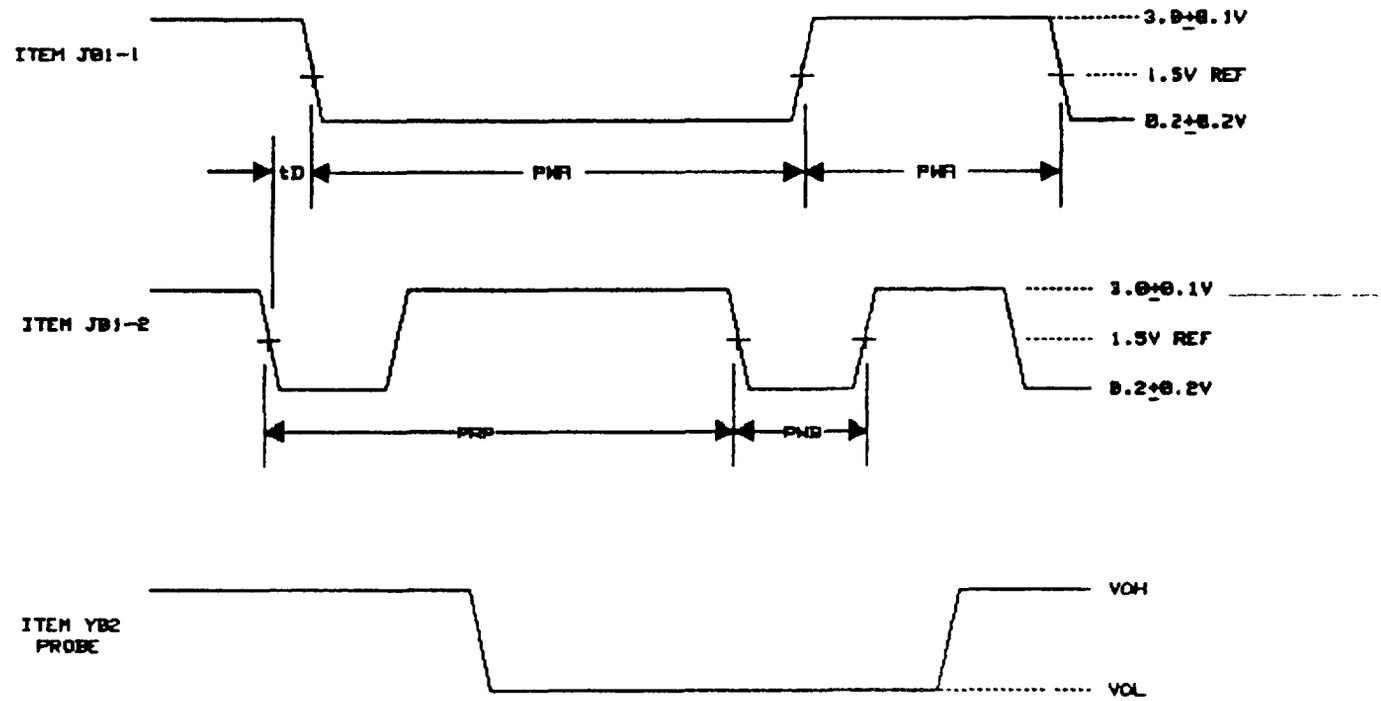

NOTE:

1. R1 = R2 = 50 Ohm ± 1 percent, 1/2W.

FIGURE 17. Timing test circuit for table XVII.



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NOTES:

1. tD = 50 ± 5 nS.
2. PWA = PRP = 10 ± 1 uS.
3. PWB = 100 ± 10 nS.
4. tTLH = tTHL = 15 nS max (10 percent to 90 percent).

FIGURE 18. Dynamic waveforms for table XVII.



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